Matto Leeuwis

m.leeuwis@erasmusmc.nl

Personal website | mleeuwis | Matto Leeuwis | 0009-0009-8728-0589

Delft, Netherlands

Research Statement

I study human movement through the lens of robotics, biomechanics, and sensorimotor neuroscience. My work explores how the nervous system interacts with the body and environment to generate skilled, adaptable behavior. Using tools such as a robotic balance simulator, I investigate how humans learn altered control dynamics, ranging from small perturbations to the complete reversal of postural control. In parallel, I use musculoskeletal models to simulate and measure the energetic cost of human posture and gait, aiming to determine whether simple models accurately describe human movement and physiology. In other applied work, I developed a dynamic model of a wheelchair that could be pushed from the side for improved social interaction. By examining motor control from a dynamic systems perspective, I aim to uncover fundamental principles in biological or engineered systems that enable us to move with intelligence and purpose.

Research interests: Cost functions that define human movement, Model-based hypotheses for how the brain realizes intelligent control, Morphological and reservoir computing applied to sensorimotor neuroscience, Biomechanical/Multibody simulations, Learning and adaptation in biological and engineered systems.

Education

- **PhD Sensorimotor Neuroscience** *Feb 2022 - Expected 2027*
  Erasmus University Rotterdam (Erasmus MC) Rotterdam, Netherlands [🌐]
  - Dissertation (in preparation): Optimality of the human balance controller: the biomechanical and neural principles that define how we stand.
  - Supervision: Patrick Forbes, Maarten Frens

- **MSc Mechanical Engineering (BioMechanical Design)** *Sep 2019 - Jan 2022*
  Delft University of Technology Delft, Netherlands
  - Grade: 8.1
  - Thesis: The Dynamics of Non-upright Castor Wheels and Their Application for Pushing Wheelchairs from the Side
  - Supervision: Heike Vallery
  - Work contributed to the founding of the startup Ziggy Mobility and continued after graduation, resulting in a PLOS ONE paper and a patent.

- **BSc Exchange Semester** *2018 - 2019*
  University of Michigan - Shanghai Jiao Tong University Joint Institute Shanghai, China
  - Grade: 9.0 (Cum Laude)
  - Courses on engineering optimization and biomedical instrumentation design.

- **BSc Mechanical Engineering** *2016 - 2019*
  Delft University of Technology Delft, Netherlands
  - Grade: 8.0 (Cum Laude)

Academic Appointments

- **Visiting Researcher** *Aug 2026 - Current*
  University of British Columbia, Sensorimotor Physiology Lab Vancouver, Canada [🌐]
  - Host: Jean-Sébastien Blouin.
  - Analyze and collect high-density surface and indwelling EMG data.
  - Simulate learning of transformed balance control.
  - Implement muscle energetics in reinforcement learning model.

- **PhD Candidate** *Feb 2022 - Jul 2026*
  Erasmus MC, Sensorimotor Neuroscience and Biorobotics Lab Rotterdam, Netherlands [🌐]
  - Design human subject experiments using a robotic balance simulator and various methods such as optical motion capture, electric vestibular stimulation, and indirect calorimetry.
  - Write scientific papers (PNAS, PLOS Comp. Biol., J. Appl. Physiol.) and present work at various conferences on neuroscience and movement control.
  - Teach and supervise students at internship, BSc, and MSc levels.
  - For a list of projects (including unpublished ones), see my website.

Publications

Jansen, Daphne N. R., Lucas Mensink, **Leeuwis, Matto**, and Patrick A. Forbes. "Cortical responses to balance perturbations persist without active postural control". In: *The Journal of Physiology* (2026). URL: https://doi.org/10.1113/JP290280.

Leeuwis, Matto, Ralf Rienks, Maarten A. Frens, Johan J.M. Pel, and Patrick Forbes. "Energetic cost of weight shifting and asymmetric weight-bearing during standing in humans". In: *Journal of Applied Physiology (in press)* (2026). URL: https://doi.org/10.1152/japplphysiol.00294.2026.

Leeuwis, Matto, Nikki Van Aerts, Ajay Seth, and Patrick A. Forbes. "The energetic cost of human standing balance and gait initiation over a range of natural postures". In: *PLOS Computational Biology* 22.7 (2026), e1013522. URL: https://doi.org/10.1371/journal.pcbi.1013522.

Mensink, Lucas, Daphne N. R. Jansen, Monika Moraczewska, **Leeuwis, Matto**, Johan Pel, and Patrick A. Forbes. "The balance N1 attenuates but persists during self-initiated and repeated perturbations, even without active balance control". In: *Journal of Neuroscience* (2026). URL: https://doi.org/10.1523/JNEUROSCI.1028-26.2026.

Leeuwis, Matto, Job Sesink, Lucy Bennett, Emma Schippers, Marieke W. Vassen, Bram T. Sterke, Katherine L. Poggensee, and Heike Vallery. "A slanted castor wheel enables pushing manual wheelchairs from the side to improve social interaction". In: *PLOS ONE* 19.9 (2024), e0307759. URL: https://doi.org/10.1371/journal.pone.0307759.

Leeuwis, Matto, Yomna Asar, Joshua J. White, Brandon G. Rasman, and Patrick A. Forbes. "Different mechanisms of contextual inference govern associatively learned and sensory-evoked postural responses". In: *PNAS (Proceedings of the National Academy of Sciences USA)* 121.32 (2024), e2404909121. URL: https://doi.org/10.1073/pnas.2404909121.

Patents

Leeuwis, Matto, Lucy Bennett, and Job Sesink. "Wheelchair". NL2033392B1. 2024. URL: https://worldwide.espacenet.com/patent/search/family/084331279/publication/NL2033392B1?q=pn%3DNL2033392B1.

Conference Presentations

Leeuwis, Matto, Lucas Mensink, Ben Rupprecht, Chris J. Dakin, Jean Sebastien Blouin, and Patrick A. Forbes. "Can humans stand in inverted control? Learning of a non-biomimetic postural task". In: *Unconventional Robots Symposium*. 2025.

Leeuwis, Matto, Nikki van Aerts, Ajay Seth, and Patrick A. Forbes. "Metabolic energy expenditure in quiet standing and gait initiation is similar over a wide range of postures". In: *International Society of Posture and Gait*. 2025.

Leeuwis, Matto, Lucas Mensink, Jean-Sébastien Blouin, Chris J. Dakin, and Patrick A. Forbes. "Reversing postural control". In: *Dynamic Walking*. 2025. URL: https://www.youtube.com/watch?v=RU0ZwT5cJbQ.

Leeuwis, Matto, Yomna Asar, Joshua J. White, Brandon G. Rasman, and Patrick A. Forbes. "Human sensorimotor systems assess contextual relevance in parallel". In: *NWO Life*. 2024.

Leeuwis, Matto, Nikki van Aerts, Ajay Seth, and Patrick A. Forbes. "Optimality and energy expenditure in standing and walking". In: *European Motor Control Meeting*. 2024.

Leeuwis, Matto, Ajay Seth, and Patrick A. Forbes. "Musculoskeletal modeling of static standing balance suggests that metabolic costs are not minimized". In: *Neural Control of Movement*. 2023.

Teaching and supervision experience

- **Course development for NeuroRobotics (Research Master Neuroscience)** *Apr 2026*
  Erasmus MC, Mario Negrello [🌐]
  - Contribute to the development of the first edition of the NeuroRobotics course for the Research Master Neuroscience.
  - Design and teach the introductory lectures on Kinematics, Dynamics, Control, and Optimization, and lead workshops on building simple robotic systems.

- **Teaching Assistant for Rigid Body Dynamics (BSc Mechanical Engineering)** *Sep 2020 and Sep 2021*
  TU Delft, Heike Vallery [🌐]
  - Full-time teaching assistant for the second-year Mechanical Engineering course Rigid Body Dynamics for the academic years of 2020/2021 and 2021/2022.
  - Develop Python code for automatically scheduling and grading weekly on-site tests for >600 students.

- **Teaching for various Physiology courses (BSc Medicine)** *2022-2026*
  Erasmus MC, Johan Pel, Jos van der Geest, Patrick Forbes, and others
  - Guide practicals for various courses, such as Exercise physiology, Blood circulation, Biomechanics, Echo, and Eye movements.

- **Mentor for Musculoskeletal Modeling and Simulation (MSc Mechanical Engineering)** *2023-2024*
  TU Delft, Ajay Seth
  - Mentor for eight student groups on proposed project "The effect of subject-specific parameters on standing balance outcomes."

- **Guest lectures**
  Erasmus MC and TU Delft
  - Erasmus MC, NeuroAI: The Computing Brain Minor 2024 and 2025 (Organizer: Mario Negrello): What does it mean to move?
  - TU Delft, Programming Foundations 2024 and 2025 (NB1035, Organizer: Nina Šoštarić): Automating experimental analysis using scientific programming
  - Erasmus MC, Research Master Neuroscience 2025 (Organizer: Gerard Borst): Introduction to the vestibular system
  - Erasmus MC, Nanjing University (NJU) Summer School 2024 (Organizer: Zhenyu Gao): Optimality and energy expenditure in standing and walking
  - 4 Internal department talks for the Erasmus MC Neuroscience Department

- **Supervision**
  Erasmus MC and TU Delft
  - 6 MSc projects
  - 2 Internship projects
  - 2 Honors projects
  - 3 BSc Final projects

Professional experience

- **Lead Experimentation for Development of Mechanical Ventilator** *May 2020 - Aug 2020*
  Project Inspiration and Delft University of Technology Delft, Netherlands [🌐]
  - Work on the development of rapidly manufacturable ventilator system for COVID-19 patients with impaired breathing.
  - Test the device according to the UK emergency ventilator standards.
  - Design modification for the device to reduce peaks in inspiratory pressure.
  - Supervision: Heike Vallery, Gerwin Smit

- **Student Engineer** *Apr 2019 - Feb 2020*
  Tromp Bakery Solutions Gorinchem, Netherlands [🌐]
  - Inspect production lines and CAD models to document potential hazards and proposed solutions following ISO norms.
  - Perform statics calculations to automatically recommend gas spring parameters for production lines.

- **Student Computer Aided Design Drafter** *Aug 2017 - Sep 2017 & Mar 2019 - Jan 2020*
  Nidec Netherlands Sliedrecht, Netherlands [🌐]
  - Create assembly manual using SolidWorks models for high-power freestanding drives.
  - Create CAD drawings and animations of multi-motor drive solutions (electric or electric-diesel hybrid) for maritime vessels.

Academic service

- **Peer Reviewer**
  Various journals
  - NPJ Robotics: 1
  - PLOS ONE: 1
  - Royal Society Interface: 1 (co-reviewed)

- **Conference Moderator** *Jul 2021*
  IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM 2021) [🌐]
  - Moderator for virtual parallel sessions; introduced speakers and facilitated general discussion after presentations.

Honors and Awards

- **Finalist 3-Minute Thesis competition 2026** *Jun 2026*
  International Society for Posture and Gait Research [🌐]
- **Editor's choice** *May 2025*
  The Journal of Physiology [🌐]
  - Our paper was selected as Editor's choice and featured on the cover of The Journal of Physiology (Volume 604, Issue 10).

- **Finalist Young Medical Delta Thesis Awards 2022** *Oct 2022*
  Young Medical Delta [🌐]
- **Fonds Bemolt Scholarship 2018** *Sep 2018*
  Stichting Fonds Bemolt [🌐]
  - Awarded to excelling students from Dordrecht pursuing scientific education abroad.

Media Coverage

- **How we maintain balance turns out to be more complex than thought** *Aug 2024*
  Featured on Amazing Erasmus MC [🌐]

Skills

- **Experimental Methods:** Human-in-the-loop robotics experiments, Optical motion capture (Qualisys, Vicon), Metabolic analysis (Indirect calorimetry; COSMED, ViaSys), Electromyography (high-density surface EMG, Myomatrix indwelling wire electrode, motor unit decomposition), Electrical vestibular stimulation (EVS/SVS), Electroencephalography (EEG; TMSi), Force plate (AMTI, Kistler), Classical conditioning, Dutch and Canadian research ethics proposals (METC, HREC, TCPS)
- **Modeling and Simulation:** Musculoskeletal modeling (OpenSim, Scone), Multibody dynamics, State-space modeling, Optimal control (e.g., LQR, Kalman filter), Numerical optimization (e.g., particle filter, genetic algorithm), Muscle energetic models, Vehicle/tire dynamics
- **Programming and Analysis:** MATLAB, LabVIEW, Python, BASH, Version control (Git, Markdown), EEGLab, Scientific computing (numerical optimization, signal processing, real-time computation), Statistics (Frequentist and Bayesian)
- **Design and Prototyping:** Mechanical Engineering, Iterative design, CAD (SolidWorks, GrabCad), 3D Printing (Cura, Ender 3, Ultimaker), Freehand sketching
- **Scientific Communication:** Scientific writing (LaTeX, Word), Figure creation (Affinity Designer/Photo, Adobe Illustrator/Photoshop, Inkscape, MATLAB)
- **Project Management and Knowledge Tools:** Course management (Brightspace, Canvas, FeedbackFruits), Reference management, Obsidian, Notion, Overleaf

Additional Information

Languages: Dutch, English, MATLAB
Personal interests: Martial arts (Secretary for Board 30 of Delftse Budo Sport Vereniging D.S.B.V. Yoroshi, 2019/2020), Bouldering, Tinkering (3D printing, Arduino, Desktop computer), Video game physics and design.